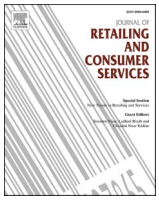
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Can you sense without being human? Comparing virtual and human influencers endorsement effectiveness

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ABSTRACT

As AI increasingly permeates digital spaces, virtual influencers are operating in similar ways as human influencers. However, given their recent introduction, the effectiveness of virtual influencers is unclear, and there is limited managerial guidance regarding when and how they should be used. Across four studies, including an empirical investigation using secondary data analysis and three scenario-based experiments spanning two cultural contexts, we find that virtual influencers are less effective than human influencers as endorsers in terms of brand attitude and purchase intention. Furthermore, we identify perceived sensory capability and credibility as the serial mechanism. Lastly, we introduce the salience of sensory cues within the ad as a theoretically important and managerially relevant moderator, which attenuates the negative effect of virtual influencers. This research extends virtual influencer literature by focusing on business-related outcomes. Our theoretical model highlights essential differences in advertising outcomes between human and virtual influencers and provides important managerial implications.

1. Introduction

Influencer marketing has proven to be an effective strategy, as exemplified by the impressive return for brands – $6.50 in revenue for each $1 spent (Tafesse and Wood, 2021; Digital Marketing, 2020). As such, it has become a ubiquitous and prioritized promotion tool for businesses (Adtraction, 2020). The global influencer marketing industry is estimated to reach $90,239 million by 2030 at a CAGR of 30.81% (PR Newswire, 2022). The increasing size and growth of influencer industry have made it an attractive space for technological innovation. The rapid advancement of artificial intelligence (AI) has given rise to a new type of influencer – the virtual influencer. Virtual influencers, defined as "3D, computer-generated personalities" (Miao et al., 2022; Deng and Jiang, 2023), have gained in popularity since their introduction in 2016 (Lou et al., 2022; Miao et al., 2022; Kamoonpuri and Sengar, 2023). Their human-like appearance and behavior allow consumers to interact with them in a manner similar to that of a human influencer, but with the potential for brands to circumvent concerns resulting from the unpredictability of associating with human influencers (e.g., scandals and ethical transgressions) (Moustakas et al., 2020; Thomas and Fowler, 2021). As the number of virtual influencers and their following grows, they can become "trusted tastemakers in one or several niches," much like human influencers (De Veirman et al., 2017; Moustakas et al., 2020). The first virtual influencer, LilMiquela, appeared on Instagram in 2016. By 2018 was named TIME magazine's most influential person on the Internet and earned more than $12 million per year endorsing brands such as Prada, Balenciaga, and Kenzo (Time, 2018). Despite the growing popularity and increased potential of virtual influencers, the effectiveness of their endorsement compared to human influencers remains unknown. For instance, 61% of consumers believe that authenticity is the primary appeal of human influencers (MarketingWeek, 2019). Another study conducted in Singapore reported that 60% of respondents found virtual influencers did not improve sales as they were not seen as "real" (Hirschmann, 2020).

Prior literature about virtual influencers primarily focuses on consumer attitudes, with little attention paid to marketing outcomes for brands. Research findings regarding the effectiveness of virtual versus human influencers have also been inconsistent. On the one hand, consumers are more likely to interact with (e.g., like and comment) virtual influencers' content (Arsenyan and Mirowska, 2021). On the other hand, despite this increased engagement, consumers tend to react more negatively and feel increased anxiety toward virtual influencers (Sands

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et al., 2022; Thomas and Fowler, 2021). To date, very little is known about how the use of virtual influencers as brand endorsers impact business outcomes, such as brand endorsement effectiveness. With this limited focus, the existing literature fails to provide theoretical guidance and managerial insights regarding both if and subsequently when virtual influencers are more effective than human influencers.

Given the human-like nature of virtual influencers, understanding the dimensions on which human and virtual influencers differ from consumers' perceptions is critical and provides insights into brand strategies for using virtual influencers as endorsers. Since virtual influencers are believed to be designed and built by human but powered by AI technology, consumers are likely to be intrigued and disturbed by virtual influencers' ability to engage in interpersonal interactions (Adtraction, 2020; Miao et al., 2022).

It has been established in the literature that consumers are often unable to distinguish real people from AI applications (Adtraction, 2020; Miao et al., 2022). Despite their visual similarities, virtual and human influencers differ in cognitive abilities. Using mind perception theory, we examine how consumers' awareness of virtual influencers' minds serves as a basis for evaluating how they interact with virtual influencers and evaluate the ads being promoted. Specifically, we compare endorser effectiveness between humans and virtual influencers by focusing on their mind perception in terms of sensory capability (e.g., hear, see, smell, touch, and taste). As sensory capability is the basic support for endorsement credibility and mind perception (Li et al., 2023; Liu and Wei, 2019), we identify sensory capability as the primary underlying mechanism which explains differences in endorsement effectiveness in terms of brand attitude and purchase intention.

To better understand the underlying mechanism, our theoretical model builds on virtual influencer literature and celebrity endorsement literature to propose a moderated serial mediation model. Given sensory-related information is of significant, we seek to validate our model by introducing sensory-related variables from the content-related perspective as an important and managerially relevant moderator (Cascio Rizzo et al., 2023). Our overall moderated serial mediation model expands current knowledge of virtual influencers and provides directions for businesses as to how to effectively use virtual influencers as brand endorsers.

2. Literature review and hypothesis development

2.1. The evolution of endorsement

Celebrities have been an effective endorsement strategy, whose fame and impacts can empower endorsed brands and promote positive brand-related endorsement outcomes (El Hedhli et al., 2021). The persuasiveness of the information depends largely on their credibility perception, which refers to the usefulness of the information in terms of expertise, attractiveness, and trustworthiness (Wiedmann and Von Mettenheim, 2020). With the rise of social media, ordinary individuals and celebrities have increasingly utilized social media as a platform to share opinions, experiences, and recommendations to attract followers, commonly referred to as influencers (Dinh and Lee, 2021; Torres et al., 2019). Influencers have a high degree of individuality in terms of unique personalities, public images, expertise, and backgrounds. Their high levels of follower engagement enable brands to strategically collaborate with individuals to achieve specific performance objectives. Accordingly, influencers can be defined into five distinct categories: celebrity influencers, mega influencers, macro influencers, micro influencers, and nano influencers (Campbell and Farrell, 2020). Existing research has investigated the effectiveness of different content-related factors (e.g., advertising disclosure, narrative strategy, and influencer-product congruence) used by different types of human influencers (Evans et al., 2017), demonstrating that influencers' credibility (i.e., trustworthiness, expertise, and attractiveness) are the major mechanisms in enhancing their effectiveness in terms of brand attitude, and purchase intention (Wang and Scheinbaum, 2017).

Recent advances in AI have introduced a new role as influencers to incorporate with brands in endorsement, the virtual influencer (also called AI influencer, Wirtz et al., 2018; Miao et al., 2022). This virtual influencer can resemble humans in many aspects (Sands et al., 2022; Thomas and Fowler, 2021) in terms of physical and social activities. The application of AI technology in a familiar social media context has been shown to generate mixed consumer attitudes toward virtual influencers. For example, previous research found that while consumers are just as likely to follow a virtual influencer compared to a human influencer, they have more negative reactions and trust the virtual influencer less (Arsenyan and Mirowska, 2021; Sands et al., 2022).

While existing research on influencers has primarily examined the factors and mechanisms associated with influencers, most of the focus remains on human influencers. The research that does examine consumers' responses to virtual influencers examines outcomes such as perceived trust and interaction intention (e.g., intention to like, comment, and follow); these outcomes remain focused on the influencer, not the endorsed brand. As far as we know, Thomas and Fowler (2021) is the only paper that investigates the endorsement effectiveness of virtual influencers. But their investigation is limited to the context of service failures by focusing on the endorser-brand relationship instead of consumer-related brand outcomes. One factor that has been examined extensively in both celebrity and influencer endorsement literature is credibility as an antecedent to endorsement effectiveness (Halder et al., 2021). However, little is known about the differences between virtual and human influencers and the consequences on endorsement outcomes such as brand attitudes and purchase intentions (see Table 1 in Appendix A for selected literature on virtual influencers).

2.2. Comparing the sensory capability of human and virtual influencers

Much like human beings, virtual influencers possess anthropomorphic appearances and the ability to interact in real-time (Hanus and Fox, 2015). While the boundary between humans and AI is becoming less distinguishable, AI applications are still perceived as machines and not human beings (Go and Sundar, 2019). One reason consumers are hesitant to interact with AI is their perception of machines' lacking mind-based characteristics (e.g., perceived professionalism, empathy, and cognitive-affective processing).

Introduced by Gray and Wegner (2012), mind perception theory proposes that people perceive the presence of another's mind before determining what someone is thinking or feeling (Gray et al., 2007; Epley and Waytz, 2010). Furthermore, mind perception consists of two distinct components: agency and experience (Gray et al., 2007; Li et al., 2023). The agency is the perceived capacity to act, whereas experience refers to the perceived ability to feel and sense. While humans' agency and experience vary across life stages and between individuals, this awareness of another's mind extends to other beings and objects, such as animals and robots (Epley and Waytz, 2010). Perceptions of another object's or being's mind are formed by assessing the object's anthropomorphic outlook and ability to experience (Stafford et al., 2014). According to mind perception theory, compared to agency, the ability to experience is implicitly perceived as being more essential to humans (Gray and Wegner, 2012). And people more strongly associate the "human mind" with experience-related terms than with agency-related terms (Gray and Wegner, 2012). In contrast to humans, the capacity of AI applications to experience senses and conscious emotions (such as shame, pride, contempt, sympathy, pain, or pleasure) is often questioned or unclear to consumers (Epley and Waytz, 2010).

Applying mind perception theory to the context of virtual influencers, AI applications possess a high degree of agency (e.g., they are capable of doing tasks autonomously). However, since they are not real human beings, their capability to experience is frequently doubted by consumers. This perception of virtual influencers' inability to feel and sense is apparent in interactions where humans challenge their true

nature and question whether they are "real" or "robots" (Laukkanen et al., 2022). Furthermore, the "word-of-machine" effect provides additional evidence that AI applications are doubted based on their inability to experience the five senses (Longoni and Cian, 2022; Luo et al., 2019). Accordingly, we propose that virtual influencers are perceived to lack genuine human mental capacities of sensory feeling as other AI agents (Gray and Wegner, 2012). Since limited research has compared human and virtual influencers, the sensory capability of virtual versus human influencers has not been examined in previous research.

2.3. Credibility of virtual influencers

Mind perception theory posits that both agency and experience are necessary for an individual to be viewed as credible (Liu and Wei, 2019). Therefore, if a being or object is perceived as lacking in mind, its credibility will be low. For example, prior research has found that when chatbots are viewed as lacking in mind, consumers trust them less (Rapp et al., 2021). Credibility, defined as "possessing expertise relevant to the communication topic such that a source can be trusted to give an objective opinion on the subject" (Goldsmith et al., 2000), consists of three factors, expertise, trustworthiness, and attractiveness (Ohanian, 1990). Expertise refers to the level of relevant knowledge, trustworthiness refers to honesty, integrity, and believability, and attractiveness refers to the pleasing level of facial features (Wiedmann and Von Mettenheim, 2020). While these three dimensions were initially applied in measuring celebrity credibility, researchers have gradually expanded their applications to various endorsement sources to evaluate the effectiveness of endorsements (Argyris et al., 2021; El Hedhli et al., 2021).

In the context of virtual influencers, their expertise may be equivalent, if not higher, than human influencers, given their objectivity, complete reliance on advanced technology, and a high degree of automation. As such, they may also be perceived as more trustworthy in providing reliable, accurate, and timely information (Lou et al., 2022). Then, in terms of trustworthiness, if trustworthiness depends on perceptions of shared or understood feelings, trustworthiness may be lower for virtual influencers compared to humans. Lastly, virtual influencers resemble the appearance of human influencers, which should enjoy similar attractiveness (Arsenyan and Mirowska, 2021). However, without the capacity to establish a human-to-human connection, which requires thinking and feeling, the credibility of virtual influencers could be harmed through this factor even though their resemblance in appearance can be really high.

The impact of this deficiency is magnified in the context of e-commerce. In influencer endorsements, influencers' credibility is established based on the information they provide (Belanche et al., 2021; Kim and Kim, 2021b). Lack of tangibility in e-commerce restricts consumers' ability to obtain a full range of information about the product (Schultz et al., 2016). Influencers then, as a marketing communication tool, satisfy consumers' need for information by compensating for these limitations of online shopping, specifically by being able to interact with a physical product. Accordingly, sensory information is an important type of information influencers can offer to consumers (Laukkanen et al., 2022; Minton et al., 2021). Sensory information pertaining to how a product looks, smells, feels, and potentially tastes compensates for the information asymmetry of online shopping and plays an increasingly significant role in shaping consumers' engagement and purchase intentions (Voorveld, 2019)—as such, integrating the five senses (sight, smell, sound, taste, and touch) in an endorsement provides a more authentic consumer experience.

As consumers treat product experiences as beneficial contexts, the richness and reliability of the sensory-related information play a pivotal role in shaping their perception of influencers' credibility (Daniel et al., 2018; Kim et al., 2021). This need for information transcends the type of influencer, human or virtual, which poses a challenge for virtual influencers. Since consumers view virtual influencers as having less sensory capability than human influencers, any sensory information provided in a virtual influencer's endorsement undermines their credibility, which in turn decreases endorsement effectiveness. Thus, in the current research, we posit that the sensory capability of the influencer will be a predictor of consumer perception of the influencer's credibility and endorsement effectiveness for brands. As such, we formally hypothesize the following.

Hypothesis 1. Virtual influencers will have lower endorsement effectiveness than human influencers.

Hypothesis 2. The effect of the type of influencer (virtual versus human) on endorsement effectiveness is serially mediated by perceived sensory capability and credibility.

2.4. The moderating role of sensory cue salience

To further validate our theory, we examine boundary conditions in which the above-mentioned effects can be attenuated. Specifically, we suggest that consumers' attention to sensory capability can be impacted by other sensory-related information, sensory cues that the endorsed brand can provide are a major part of the ad content concerning sensory (Cascio Rizzo et al., 2023). Sensory cues encompass signals related to the five senses, which have been shown to increasingly impact consumers' product evaluations (Hultén, 2012; Li et al., 2023). For instance, sensory cues containing physical product attributes, such as taste or smell, affect consumers' quality perception, product attitudes, consumer experiences, and affective responses (Agapito et al., 2013; Helmefalk and Hultén, 2017; Li et al., 2019). While sensory cues provide additional information to consumers, their importance and relevance differ across product categories. In the present investigation, we operationalize sensory cue salience in the endorsement content by manipulating the presence of ad content related to the five senses within an influencer's endorsement. We predict that the sensory cue salience will direct consumers' focus to the virtual influencers' sensory capability, or lack thereof, which will impact endorsement effectiveness. When the sensory cue is salient, consumers' sensory capability perception toward virtual influencers will heighten and be reinforced, and their endorsement effectiveness regarding lack of sensory capability will be reduced. While consumers perceive the high sensory capability of human influencers, the level of sensory cue salience will not provoke consumers' doubts regarding sensory capability and endorsement effectiveness. When the sensory cue is not salient, consumers' focus on sensory-relevant information will be reduced and shifted to other perspectives of the brand. Accordingly, the perceived level of sensory capability from the type of influencer will not impact endorsement effectiveness significantly.

Given the potential benefits of virtual influencers, identifying circumstances when they are suitable for brand endorsement is managerially relevant. Therefore, if the endorsement content can be altered based on the type of influencer to include different types of information, consumers may be more receptive to virtual influencers. Thus, we propose the following hypothesis.

Hypothesis 3. Sensory cue salience moderates the relationship between the type of influencer and the effectiveness of endorsement through perceived sensory capability.

3. Methodology

In this research, a multimethod design consisting of four studies combines secondary data exploration and experimental methods, to provide evidence for the proposed hypotheses. In Study 1, we seek to explore evidence for sensory capability perception and endorsement effectiveness differences between human and virtual influencers. Two studies (Studies 2a and 2b) further demonstrate the serial mechanisms while ruling out other alternative mechanisms. Finally, Study 3 extends

the investigation by examining the moderating role of sensory cue salience.

3.1. Study 1

Study 1 seeks to provide real-world evidence comparing the perceptions of sensory capability and endorsement effectiveness between human and virtual influencers. For three reasons, we selected Instagram as the context for this first empirical investigation. First, Instagram is a highly visual platform, emphasizing photos and videos. Second, Instagram is the first platform and the primary aggregation site for virtual influencers, home to 35 verified virtual influencers in 2022 (VirtualHumans, 2022). Lastly, Instagram is one of the fastest-growing social network sites, with 1.3 billion monthly active users making it an attractive platform for influencer campaigns (InfluencerMarerktingHub, 2022). Therefore, Instagram provides a perfect context with enough observations to examine how consumers interact differently with virtual versus human influencers.

3.1.1. Data collection

We utilized Python to scrape posts and public comments from two selected influencer endorsers on Instagram. The two influencers chosen for this study are LilMiquela[1] as the virtual influencer and Marta Cygan[2] as the human influencer. The selection criteria for these two influencers were as follows: First, as LilMiquela is the first virtual influencer to acquire mainstream recognition (Data Conomy, 2022) and receives top-rank followers in 2021 (Influencity, 2021), we select LilMiquela as the virtual influencer representative. Second, six human influencers were selected based on their match with LilMiquela in terms of age, category, and residence by a specialist team and further rated by 50 participants in terms of their similarity (i.e., personal style, post content, and followers) with LilMiquela. Marta Cygan was finally chosen as the human influencer representative who shares the highest similarity with LilMiquela. Given the rapid growth of virtual influencer audiences, to minimize any confounding factors, we limited our study to posts and comments made between January 1, 2021, and April 30, 2022, which was the start of the data collection process for this study. We collected both influencer posts and corresponding comments during this period from followers and non-followers alike to avoid any possibility of biases resulting from parasocial relationships, familiarity with the influencer, or propensity to interact favorably stemming from following the influencer. Overall, we collected a total of 466 influencer posts and 107,218 corresponding comments, with 228 posts and 97,112 comments for the virtual influencer and 238 posts and 10,106 comments for the human influencer.

3.1.2. Results

The level of social media engagement is considered a crucial indicator in assessing endorsement effectiveness (Ren et al., 2023), we first conducted an analysis to compare consumers' engagement with influencers' posts in terms of likes and comments (Coelho et al., 2016), and then analyzed the content within consumer comments using linguistic analysis that specifically focuses on consumers' sensory perception of influencers.

First, we conducted ANOVA analysis to compare consumers' engagement in liking and commenting to determine whether consumers' engagement with virtual (VI) versus human influencers (HI) differed. From the analysis, virtual influencers result in more likes (M_{VI} = 76,030.82, SD = 88,166.67; M_{HI} = 2,907.62, SD = 1,908.70; F (1, 464) = 163.65, $p < 0.001$), comments (M_{VI} = 622.81, SD = 1,277.48; M_{HI} = 43.76, SD = 29.78; F (1, 464) = 48.88, $p < 0.001$), than human influencers.

However, while comments and likes are common measures of engagement, the text within the comments can be analyzed to provide additional information regarding consumers' thoughts and feelings. To conduct text analysis, we relied on LIWC-22, a widely applied linguistic analysis software that captures quantitatively the frequency of psychological states, including emotions, perceptions, and social relationships in text (LIWC2022, Boyd et al., 2022). LIWC provides categories of words, three of which are related to our hypotheses: *mind perception* (experience, agency, and overall mind perception), *perception* (visual and auditory), and *affect*(positive emotion, negative emotion, anxiety, and swear words) (Table 1 summarizes the measured variables).

From the analysis, we find that virtual influencers receive significantly more comments than human influencers both on experience category (M_{VI} = 0.04, SD = 0.20; M_{HI} = 0.02, SD =0.15; F (1, 107216) = 79.29, $p < 0.001$), agency category (M_{VI} = 0.08, SD = 0.27; M_{HI} = 0.04, SD = 0.20; F (1, 107216) = 179.27, $p < 0.001$), and overall mind perception category (M_{VI} = 0.11, SD = 0.31; M_{HI} = 0.06, SD = 0.24; F (1, 107216) = 247.41, $p < 0.001$). In the perception category, virtual influencers receive significantly more comments on visual perception (M_{VI} = 1.41, SD = 6.51; M_{HI} = 1.15, SD = 6.86; F (1, 91416) = 10.41, $p < 0.001$), and auditory perception (M_{VI} = 0.17, SD = 2.43; M_{HI} = 0.10, SD = 2.70; F (1, 91416) = 4.03, $p < 0.05$) than human influencers. This initial result illustrates that consumers talk more about mind perception of virtual influencers than human influencers.

However, when we conduct further analysis of the emotional orientation of these comments, we find that virtual influencers receive significantly less positive emotion (M_{VI} = 3.56, SD = 12.04; M_{HI} = 5.96, SD = 18.25; F (1, 91416) = 226.80, $p < 0.001$) and higher negative emotion (M_{VI} = 0.72, SD = 5.38; M_{HI} = 0.12, SD = 2.50; F (1, 91416) = 83.87, $p < 0.001$), anxiety (M_{VI} = 0.19, SD = 3.07; M_{HI} = 0.01, SD = 0.51; F (1, 91416) = 22.85, $p < 0.001$), and swear (M_{VI} = 1.04, SD = 7.40; M_{HI} = 0.12, SD = 2.86; F (1, 91416) = 102.93, $p < 0.001$) than human influencers. This suggests that although consumers seem to be more engaged with virtual influencers through the number of likes and comments, these comments exhibit negative valance through words such as worry, fear, and swear. Hence, we provide initial some evidence in support of H1.

3.2. Study 2a

While Study 1 provided empirical evidence for our hypotheses, Study 2a and 2b are intended to provide additional support for the main effect proposed in H1 and the mediation effects proposed in H2. Study 2a sought causal evidence for our proposed main and mediation effects by testing whether influencer type (virtual influencer vs. human influencer) can impact endorsement effectiveness (measured by brand attitude and purchase intention) through perceived sensory capability and credibility.

3.2.1. Study design

A single factor (influencer type: virtual vs. human influencer) between-subjects experiment was conducted. The study was conducted via a Chinese online crowdsourcing platform Credamo, similar to MTurk in April 2022. Two hundred and sixty-one participants (age range = 18–60, Mage = 30.32, 39.46% male) were selected after deleting nine participants who did not pass the attention check. Each participant received a nominal payment in exchange for participation. After consenting to participate, each respondent was randomly assigned to one of two experimental conditions (human vs. virtual influencer). The participants were asked to imagine that they were browsing social media and found an influencer's home page showing their name, image, brief introduction, and recent updates. We manipulated the influencer condition by manipulating their name and description, such that in the virtual influencer condition, the influencer called "Virtual_Namey" was introduced in a brief description as a "virtual influencer_fashion blogger"; while in the human influencer condition, an influencer called

[1] https://instagram.com/lilmiquela.
[2] https://instagram.com/lifeofboheme.

Table 1
Measurement of variables.

Category		Variable names	Description	Descriptive statistics				
				VI		HI		*F*
				M	SD	M	SD	
Social media engagement		Like	Number of likes on each post.	76,030.82	88,166.67	2,907.62	1,908.70	163.65***
		Comment	Number of comments on each post.	622.81	1,277.48	43.76	29.78	48.88***
LIWC analysis	Mind Category	Experience	Whether the comments contained experience category indicating experience perception.	0.04	0.20	0.02	0.15	79.29***
		Agency	Whether the comments contained agency category indicating agency perception.	0.08	0.27	0.04	0.20	179.27***
		Overall mind perception	Whether the comments contained word relating to mind perception.	0.11	0.31	0.06	0.24	247.41***
	Perception category	Visual	Whether the comments contained word relating see, look, saw etc.	1.41	6.51	1.15	6.86	10.41***
		Auditory	Whether the comments contained word relating sound, hear, music etc.	0.17	2.43	0.10	2.70	4.03*
	Affect category	Positive emotion	Whether the comments contained positive emotion including good, love, happy etc.	3.56	12.04	5.96	18.25	226.80***
		Negative emotion	Whether the comments contained negative emotion including bad, hate, tired etc.	0.72	5.38	0.12	2.50	83.87***
		Anxiety	Whether the comments contained anxiety emotion including worry, fear, afraid etc.	0.19	3.07	0.01	0.51	22.85***
		Swear words	Whether the comments contained swear words.	1.04	7.40	0.12	2.86	102.93***

Note: *p < 0.05, **p < 0.01, ***P < 0.001.

"Namey" was introduced as a "fashion blogger." Both scenarios share the exact same image and recent updates. After reading about the influencer, we measured endorsement effectiveness including brand attitude and purchase intention, followed by perceived sensory capability and perceived credibility. The participants were then asked to complete a manipulation check. The original English measurement scales were translated into Chinese and back into English by professional translators to ensure measurement equivalence (Brislin, 1980).

3.2.2. Measurement

The manipulation check for influencers' perception was measured using one seven-point semantic-differential scale from Mozafari et al. (2021), "Please rate whether you think you talked to a virtual influencer or a human influencer" anchored by virtual influencer (1) and human influencer (7).

Perceived sensory capability was measured using a scale from Stafford et al. (2014). The scale includes items such as "I perceived capacity of the Namey/Virtual_Namey to see." Cronbach's alpha was 0.89, indicating satisfactory reliability for perceived sensory capability.

Perceive credibility was measured using a scale from Janssen et al. (2022), which included "I think Namey/Virtual_Namey is untrustworthy/trustworthy." Cronbach's alpha was 0.87, indicating satisfactory reliability for perceived credibility.

Endorsement effectiveness was measured using brand attitude and purchase intention. Brand attitude was measured using four items from Kim and Kim (2021a) with the question, "How did you feel about the brand that was endorsed by the Namey/Virtual_Namey?", on a seven-point semantic differential scale, including "Desirable/not desirable." Cronbach's alpha was 0.93, indicating satisfactory reliability for brand attitude. Purchase intention was measured using a scale from Janssen and colleague (2022) with one item, "The next time you are looking to purchase this type of product, how likely are you to buy this product?". The detailed measures are displayed in Appendix C.

3.2.3. Results

Manipulation check. The manipulation check for type of influencer is significant with participants in the virtual influencer group perceiving they are viewing virtual influencers more than those in the human influencer condition ($M_{VI} = 3.24$, $SD = 1.18$; $M_{HI} = 5.02$, $SD = 1.00$; $t = -13.21$, $p < 0.001$). Therefore, the manipulation was successful.

Sensory capability. As predicted in H2, influencers can lead to different levels of sensory capability perception. A one-way ANOVA showed that consumers perceive lower sensory capability in the virtual influencer condition than in the human influencer condition ($M_{VI} = 5.06$, $SD = 1.39$; $M_{HI} = 5.70$, $SD = 0.86$; $F(1, 259) = 20.32$, $p < 0.001$), which provides supports for H2.

Perceived credibility.A one-way ANOVA showed that consumers' perceived lower credibility in the virtual influencer group than in the human influencer group ($M_{VI} = 5.13$, $SD = 1.10$; $M_{HI} = 5.84$, $SD = 0.90$; $F(1, 259) = 32.07$, $p < 0.001$). The mediation analysis demonstrated that sensory capability (indirect effect = −0.37, SE = 0.09, 95% CI = [−0.57, −0.20]) mediated the relationship between influencer type and perceived credibility. This provides additional supports for H2.

Endorsement effectiveness. A one-way ANOVA was utilized to test for the effect of influencers' endorsement, including brand attitude and purchase intention, also supporting H2. A one-way ANOVA showed that consumers' brand attitudes were higher in the human influencer group than in the virtual influencer group ($M_{VI} = 5.23$, $SD = 1.22$; $M_{HI} = 5.76$, $SD = 0.87$; $F(1, 259) = 16.70$, $p < 0.001$). Similarly, a one-way ANOVA showed that consumers' purchase intention was higher in the human influencer group than in the virtual influencer group ($M_{VI} = 4.85$, $SD = 1.25$; $M_{HI} = 5.65$, $SD = 1.22$; $F(1, 259) = 27.55$, $p < 0.001$).

Serial Mediation: We then employed a serial mediation analysis using PROCESS macros (Hayes, 2018; Model 6 for moderated serial mediation) with perceived sensory capability and perceived credibility as sequential mediators and endorsement effectiveness as the outcome. The results revealed that respondents perceived virtual influencers to have lower perceived sensory capability and perceived credibility, resulting in lower brand attitude (indirect effect = −0.17, SE = 0.06, 95% CI = [−0.30, −0.07]) and purchase intention (indirect effect = −0.29, SE = 0.83, 95% CI = [−0.47, −0.14]; see Figs. 2 and 3 in Appendix D for path models with estimated coefficients).

3.3. Study 2b

Building on empirical evidence found in Study 2a, Study 2b extends the investigation of the influencer endorsement effect in three ways. First, to further test the proposed mechanism, we manipulated whether identity disclosure magnifies the deficiency in sensory capability perception for virtual influencers. According to our hypotheses, virtual and human influencers share similarities in physical appearance and social activities. Acknowledging consumers' hesitancy towards virtual

influencers can highlight the fact that they are not real humans, which in turn can magnify the differences in sensory capability and credibility, and further impact endorsement effectiveness. On the other hand, we expect that for virtual influencers with identity concealment, their realistic anthropomorphic appearance may lead consumers to perceive them as more similar to real humans, which attenuates the effects on perceived sensory capability, credibility, and endorsement effectiveness.

Study 2b also intends to rule out other alternative mechanisms, such as perceived similarity and perceived attractiveness, which have been shown to account for the influencer attributes impacting influencer marketing effectiveness (Masuda et al., 2022). Lastly, Study 2b is also designed to test the robustness of the effects found in Study 2a by using a different sample.

3.3.1. Study design

The study was conducted on Prolific in October 2022. Five hundred and eleven participants (age range = 18–60, Mage = 29.58, 41.49% male) were selected in exchange for a monetary reward after deleting fourteen participants who did not pass the attention check. Each participant was randomly assigned to one of three conditions (human influencer vs. virtual influencer-source disclosure vs. virtual influencer-source concealment) in a between-subjects design. The participants were asked to imagine that they were browsing social media and found an influencer. Next, participants were shown a home page of the influencer with their name, image, brief introduction, and recent updates.

In the human influencer condition, the influencer is called "Namey" who is a "fashion blogger" whereas in the virtual influencer condition with source disclosure, the influencer was called "Virtual_Namey" who is a "virtual influencer fashion blogger". Additional comments suggesting that "consumers are questioning her identity, with many challenging that she is a virtual digital being and not a human being." were provided to the participants in the virtual influencer with source disclosure condition. In contrast, in the virtual influencer with source concealment condition, the additional comments regarding the virtual influencer's identity were omitted.

After reading the endorsement information, the participants were asked to complete the measurement items for endorsement effectiveness (in terms of brand attitude ($\alpha = 0.91$) and purchase intention), perceived sensory capability ($\alpha = 0.86$), perceived credibility ($\alpha = 0.86$), and followed by manipulation checks. Perceived similarity was measured using a scale from Schouten et al. (2020) and perceived attractiveness from Masuda et al. (2022).

3.3.2. Results

Manipulation Check.The manipulation check for type of influencer is significant, with participants in the virtual influencer group perceiving they are viewing virtual influencer more than those in the human influencer group (MVI_source disclosure = 3.74, SD = 0.97; MHI= 5.61, SD = 0.92; MVI_source concealed = 4.33, SD = 1.20; F (2, 508) = 139.04, $p < 0.001$). This suggests that the manipulation was successful.

Endorsement effectiveness. For brand attitude, a one-way ANOVA revealed that the type of influencer can affect brand attitude (F (2, 508) = 14.80, $p < 0.001$). Consistent with results from Study 2a, consumers have less favorable brand attitudes toward virtual influencers versus human influencers (MVI_source disclosure = 5.20, SD = 1.00; MHI = 5.69, SD = 0.94; F (1, 323) = 20.40, $p < 0.001$). In addition, source disclosure moderated the effect of influencer type on brand attitude, such that compared to the source disclosure group, consumer brand attitude is more favorable in the source concealment group (MVI_source concealed = 5.72, SD = 0.91; F (1, 335) =23.92, $p < 0.001$), which is similar to the human influencers group (F (1, 358) = 0.07, p= 0.80).

For purchase intention, a one-way ANOVA revealed that the type of influencer can have differential effect on purchase intention (F (2, 508) = 12.18, $p < 0.001$). Consistent with results from our previous studies, virtual influencers, relative to human influencers, decreased consumers' purchase intention (MVI_source disclosure = 5.11, SD = 1.39; MHI = 5.66, SD = 1.21; F (1, 323) = 14.54, $p < 0.001$). However, source disclosure moderated the effect of virtual influencers on purchase intention; consumers had a higher purchase intention in source concealment group than source disclosure group (MVI_source concealed = 5.73, SD = 1.11; F (1, 335) = 20.66, $p < 0.001$). And purchase intention of the source concealment group is similar to the human influencers group (F (1, 358) = 0.33, p= 0.56).

Sensory capability. A one-way ANOVA revealed that influencers can lead to significantly different levels of sensory capability perception (F (2, 508) = 20.38, $p < 0.001$). Consistent with results from Study 2a, consumers' perceived sensory capability is lower for virtual influencers compared to human influencers, decreased (MVI_source disclosure = 5.11, SD = 1.27; MHI= 5.67, SD = 0.91; F (1, 323) = 21.22, $p < 0.001$). In addition, source disclosure moderated the effect of influencer type on perceived sensory capability. Consumers perceived higher sensory capability in source concealment group than source disclosure group (MVI_source concealed = 5.76, SD = 0.78; F (1, 335) = 33.40, $p < 0.001$). And the perceived sensory capability in the virtual influencer source concealed condition is not different from the human influencers condition (F (1, 358) = 1.05, p= 0.31). The results provide additional support for H1 and H2.

Perceived credibility.A one-way ANOVA showed that consumers perceived lower credibility in the virtual influencer group than in the human influencer group (MVI_source disclosure = 5.13, SD = 1.11; MVI_source concealed = 5.75, SD = 0.86; MHI=5.68, SD = 0.87; F (2, 508) = 20.34, $p < 0.001$). Consistent with results from Study 2a, virtual influencers, relative to human influencers, are perceived to have lower credibility (F (1, 323) = 24.32, $p < 0.001$). However, source disclosure moderated the effect of virtual influencers on perceived credibility. Consumers perceived higher credibility in source concealment group than source disclosure group (F (1, 335) = 32.73, $p < 0.001$). And perceived credibility for source concealment group is not significantly different from human influencers group (F (1, 358) = 0.61, p = 0.44).

Alternative Mechanisms. There was no significant difference for perceived similarity (MVI_source disclosure= 5.98, SD = 0.64; MHI=5.82, SD = 0.96; MVI_source concealed = 5.97, SD = 0.80; F (2, 508) = 1.98, p = 0.14), or the perceived attractiveness (MVI_source disclosure = 5.56, SD = 1.07; MHI = 5.46, SD = 1.17; MVI_source concealed = 5.57, SD = 1.07; F (2, 508) = 0.55, p= 0.58).

Mediation Analysis. Finally, we conducted a serial mediation analysis using PROCESS macros (Hayes, 2018; Model 6 for moderated serial mediation) with influencer type as the independent variable, perceived sensory capability as the first mediator, perceived credibility as the second mediator, and endorsement effectiveness as the dependent variable.

Since there was no difference in endorsement effectiveness between the human influencer condition and the virtual influencer-source concealment condition, it is not surprising that perceived sensory capability and perceived credibility did not mediate the effect of influencer type on endorsement effectiveness for these conditions in terms of brand attitude (indirect effect = 0.02, SE = 0.02, 95%CI = [−0.02, 0.07]), and purchase intention (indirect effect = 0.03, SE = 0.03, 95%CI = [−0.03, 0.08]).

More importantly, replicating our effects from Study 2a, the 95% confidence interval for the comparison between human influencer and virtual influencer-source disclosure did not include zero, indicating that perceived sensory capability mediates the effect of influencers on brand attitude through its effect on perceived credibility (indirect effect = −0.17, SE = 0.04; 95%CI = [−0.26, −0.10]) and purchase intention (indirect effect = −0.19, SE = 0.05; 95%CI = [−0.31, −0.11]). The path model with estimated coefficients is displayed in Appendix D (See Figs. 4 and 5).

3.4. Study 3

Given virtual influencers' shown deficiencies as endorsers, what can

businesses do to mitigate these genitive effects? Study 3 sought to further validate our serial mediation model by investigating the moderation effect of sensory cue salience by manipulating it. This study also demonstrates the robustness of our results by switching to a product category that mainly features a different type of sense, hearing.

3.4.1. Study design

Study 3 consists of a 2 (influencer type: virtual influencer vs. human influencer) x 2 (sensory cue salience: high vs. low) between-subjects design. The study was administered via Prolific in October 2022. Three hundred eighty participants (age range = 18–60, Mage = 41.50, 50.00% male) completed the experiment in exchange for a monetary reward with all passed the attention check. Headphones from a real brand, Bose, were used as the advertised product. To capture the change in endorsement effectiveness, firstly, participants were asked to rate their brand attitude and purchase intention of Bose headphones, which is a known brand. Then, each participant was randomly assigned to one of four conditions with a new product from Bose, the Bose QuietComfort Earbuds. Influencer type was manipulated by showing a photo of two real people in the human influencer condition or showing the avatar version of the same photo generated by an app in the virtual influencer condition (Virtual_Chelsea, and Virtual_Mary). Participants' brand attitude and purchase intention were measured again and were compared for subtle changes.

Sensory cue salience was manipulated by whether the product description focuses on features (low sensory salience) versus experience through different senses such as hearing and touch (high sensory salience). Specifically, In the high sensory salience condition, the product endorsement message reads, "I like the earbuds so much! These earbuds fit my ear well, and I can hear high quality sound. I feel comfortable and safe wearing the earbuds … These earbuds sound clear without noises. My ears feel comfortable even after wearing it for 6 h" In the low sensory cue salience condition, the message reads, "I like the earbuds so much! It has shark fin earcups designed to be secure and non-slip, with 11 levels of noise cancellation to adapt to different scenarios … The internal and external microphone synchronized detection of both active and passive noise cancellation ensure the sound quality." After reading the background material, the participants were asked to complete the measurement items for endorsement effectiveness, including brand attitude ($\alpha = 0.96$) and purchase intention, followed by perceived sensory capability ($\alpha = 0.98$), perceived credibility ($\alpha = 0.96$), and manipulation check, similar to those used in Studies 2a and 2b.

3.4.2. Results

Manipulation Check. The manipulation check for type of influencer is significant at $p < 0.001$, with participants in the virtual influencer group perceiving they are viewing virtual influencer more than those in the human influencer group (MVI = 1.81, SD = 1.42; MHI = 4.73, SD = 1.84; $t = -17.27$).

Perceived sensory capability. A simple effect analysis showed a significant interactive effect of influencer type and sensory cue salience on perceived sensory capability ($F(1,376) = 4.43$, $p < 0.05$). Perceived sensory capability is significant different for virtual versus human influencers both in high (MVI = 1.81, SD = 1.34; MHI = 4.96, SD = 1.47; $F(1, 376) = 149.51$, $p < 0.001$) and low sensory cue salience (MVI = 2.07, SD = 1.41; MHI = 4.61, SD = 1.41; $F(1, 376) = 245.62$, $p < 0.001$) condition.

Endorsement effectiveness.A simple effect analysis showed a significant interactive effect of influencer type and sensory cue salience on brand attitude ($F(1, 376) = 28.72$, $p < 0.001$). Under the high sensory cue salience condition, brand attitude is significantly higher in human influencers than virtual influencers (MVI = 3.23, SD = 1.31; MHI = 4.98, SD = 1.39; $F(1, 376) = 91.38$, $p < 0.001$) whereas brand attitude is not significantly different between human and virtual influencers under low sensory cue salience condition (MVI = 4.44, SD = 1.17; MHI = 4.78, SD = 1.25; $F(1, 376) = 3.20$, $p = 0.08$).

For purchase intention, a simple effect analysis showed a significant interactive effect of influencer type and sensory cue salience on purchase intention ($F(1, 376) = 13.74$, $p < 0.001$). Under high sensory cue salience condition, purchase intention is significantly higher in human influencers than virtual influencers (MVI = 2.59, SD = 1.44; MHI = 3.84, SD = 1.77; $F(1, 376) = 32.15$, $p < 0.001$). In contrast, there is no significant different in purchase intention between human influencers and virtual influencers under low sensory cue salience condition (MVI = 3.98, SD = 1.38; MHI = 4.05, SD = 1.54; $F(1, 376) = 0.11$, $p = 0.75$).

Furthermore, there is a significantly higher reduction in brand attitude (MVI = −1.74, SD = 1.57; MHI = −0.79, SD = 1.18; $F(1, 378) = 44.21$, $p < 0.001$) and purchase intention (MVI = −1.24, SD = 1.89; MHI = −0.66, SD = 1.78; $F(1, 378) = 9.59$, $p < 0.001$) for virtual influencers compared to human influencers.

Moderated Mediation Analysis.We tested the moderated mediation model using the PROCESS macros (Hayes, 2018; Model 83 for moderated serial mediation). We found significant evidence of a two-way interaction (influencer type x sensory cue salience) on brand attitude (a x b = -.61, se = 0.29, 95% CI = [−1.18, −0.04]) and purchase intention (a x b = −0.61, se = 0.29, 95% CI = [−1.18, -.04]). The results supported the indirect mediation effect of perceived sensory capability and perceived credibility on brand attitude (Low sensory cue salience: a x b = -.69, se = 0.11, 95% CI = [-.91, -.50]; High sensory cue salience: a x b = -.85, se = 0.12, 95% CI = [−1.11, -.63]) and purchase intention (Low sensory cue salience: a x b = -.55, se = 0.10, 95% CI = [-0.77, -.36]; High sensory cue salience: a x b = -.68, se = 0.12, 95% CI = [-0.93, -.45]). The path model with estimated coefficients is displayed in Appendix D (See Figs. 6 and 7).

4. Discussion

4.1. General discussion

This study aims to explore virtual influencers' impact on endorsement effectiveness and the underlying mechanism based on mind perception theory and endorsement theory. First, consistent with prior literature (Arsenyan and Mirowska, 2021), the secondary data of the current study supports the notion that consumers have a higher level of engagement with virtual influencers than human influencers. Additionally, Study 1 further provides field evidence for mind perception theory and sensory capability perception differences in consumers' perceptions of virtual and human influencers.

Second, the results show that virtual influencers are less effective as endorsers compared to human influencers. This effect is attributed to lower perceived sensory capability and perceived credibility. Besides, the robustness of the serial mechanisms is demonstrated through ruling out other alternative explanations (perceived similarity and perceived attractiveness). Given that virtual influencers, as advanced AI applications, mimic human beings, consumers tend to perceive their minds in terms of agency and experience (Epley and Waytz, 2010). Due to the nature of AI, consumers tend to perceive lower experience from virtual influencers, which leads to lower perceived sensory capability (Gray and Wegner, 2012). Their lack of sensory capability can cause consumer skepticism on the authenticity of endorsement information and decreased perceived credibility (Kim et al., 2021). However, virtual influencer's negative impact on endorsement effectiveness can be attenuated by identity concealment.

Additionally, the effectiveness of influencers can be easily used by brands through varying the focus of endorsement message content. Study 3 identified a managerially relevant boundary condition for the effect of influencer type on endorsement effectiveness. While the results demonstrated that human influencers have a significantly higher effectiveness in endorsement than virtual influencers, the endorsement effectiveness of virtual influencers can be enhanced when the sensory cue salience is low. These findings provide further validation for our theory built on mind perception theory and celebrity endorsement

theory.

Overall, the results of the current study demonstrate that although virtual influencers may receive higher engagement indicated by more comments as suggested by prior literature, as endorsers, they are less trusted (Moustakas et al., 2020; Robinson, 2020) and are not as effective (Thomas and Fowler, 2021) as human influencers. Through our theoretical and empirical investigations, our research provides both theoretical contributions and managerial implications.

4.2. Theoretical contributions

This research contributes to existing research on influencer marketing by comparing human and virtual influencers. Previous research has focused on comparing the effectiveness of sources in terms of different human celebrities and social media influencers, including macro-influencers or mega influencers, and micro-influencers (Kim and Kim, 2020). Another stream of prior research has examined content-related strategies (e.g., ad disclosure level, ad disclosure time) to enhance endorsement effectiveness. However, the prior research has been based on the premise that influencers are just like human. Little research, to our knowledge, has investigated the endorsement effectiveness of virtual influencers. Through investigating the endorsement effectiveness of virtual influencers, we document that virtual influencers decrease endorsement effectiveness in terms of consumers' brand attitudes and purchase intentions.

Second, the research investigated the psychological mechanism of virtual influencers by bridging the literature of mind perception and celebrity endorsement. Previous research on AI has shown that the role AI plays and the anthropomorphism of AI can enhance consumers' acceptance and attitude toward AI (Marikyan et al., 2022; Uysal et al., 2022). However, until recently, AI has generally been viewed and discussed as being simplistic or superficial avatars lacking behavioral realism. Little research, however, has investigated AI as a virtual influencer. The virtual influencers in our study are digital human avatars with both form and behavior realism (Miao et al., 2022). The present research addresses this gap by investigating the possible marketing outcomes of AI's new role as an influencer in the endorsement context. Furthermore, this study sheds light on the mechanism of virtual influencers by mind perception theory and cue utilization theory, suggesting the serial-mediating roles of sensory capability and credibility perception.

Moreover, this paper also bridges gaps between the literature on communication strategy and influencers. Previous research has addressed the communication strategy of influencers from content perspectives, including ad disclosure level and disclosure time (Yang, 2022). However, this study introduces the moderating role of sensory cue salience as another aspect from the content perspective. While amplifying the influencers' literature from content characteristics, this study also contributes to the communication literature by emphasizing the sensory cues in endorsement communication.

4.3. Managerial implications

Managerially, this research provides information for marketers who employ virtual influencers and suggests that virtual influencers may not be an effective alternative to human influencers for endorsement, especially for sensory-essential product categories such as beauty, lifestyle, and fashion. Compared to human influencers, virtual influencers led to a significantly lower perception of sensory capability, as well as lower perceived credibility, ultimately reducing endorsement effectiveness. It, therefore, seems advisable that enterprises do not rush to replace human influencers with virtual ones, even though virtual influencers may offer the brand additional control and cost savings.

Furthermore, this study provides a practical foundation regarding how brands might more effectively introduce virtual influencers as endorsers. We explore the impact of virtual influencers on the effectiveness of endorsement and consumers' psychological responses to virtual influencers, which can be explained by the perception of sensory capability and credibility. Our results suggest that brands should consider the level of "human" experiences needed for their brand to succeed and design the endorsement content accordingly. A virtual influencer might not be the best candidate for endorsements that are highly relevant to the senses.

Finally, our research provides practical strategies for enterprises to enhance virtual influencers' endorsement effectiveness. When the sensory cue salience is low, consumers' focus on perceived sensory capability will be reduced, reducing the impact on the perception of sensory capability. Therefore, in addition to considering which type of influencers to employ for endorsements, brands must also design endorsed content carefully. Though enriching sensory cues is important, the type of sensory cues also needs attention.

4.4. Limitations and future research

Overall, the current research takes initial steps toward a better understanding of the endorsement effectiveness of virtual influencers, consumers' psychological reactions to them, and the contrast of these effects with human influencers. However, there are nonetheless several limitations. First, this study does not involve detailed features of the virtual influencers, such as appearance (i.e., human-like vs. anime-like), attractiveness, and celebrity status, which are source-related factors that can impact consumer attitudes. Future studies should consider these characteristics' effects and differences when investigating virtual influencers.

Second, virtual influencers are a current trend in the metaverse with great potential for development and application. However, this study only explores the impact of virtual influencers' endorsement on social media platforms. Social media platforms differ in terms of user demographics, content formats, and engagement patterns, which can impact the effectiveness of virtual influencer endorsements. Future research could examine consumer reactions to virtual influencers across different social media platforms. More specifically, as live streaming becomes a major channel for endorsement, the impact of virtual influencers on endorsement effectiveness in live streaming also needs further investigation.

Furthermore, there exists a varied need for sensory cues across product types. For example, food is regarded as a product that needs more sensory cues, while sensory cues seem almost unnecessary for products like credit cards (Li et al., 2019). This study manipulates the sensory cue salience from a content perspective, and future research can investigate the sensory cue salience from a product perspective. In addition, future research can investigate the effect of virtual influencers on product characteristics and explore more communication strategies for virtual influencers.

Finally, this study employs a scenario experiment and self-report measurements at a single time point. Future research could adopt a longitudinal design and use neuroscientific methodologies to construct a dynamic evolution model to examine virtual influencer effectiveness. Moreover, as the secondary data can be disturbed by various factors and marketing outcome is hard to track, the current research focuses on social media engagement, brand attitude, and purchase intention to validate the endorsement effectiveness. Future research could establish partnerships with enterprises to constant tracking and investigate the real-world impact of virtual influencer endorsement, such as tracking revenues and building an effectiveness formula, providing more direct and tangible evidence of endorsement effectiveness.

Declaration of competing interest

The authors declare that they have no known competing financial interests that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A

Table 1
Selected Literature on Virtual Influencers

Article	Influencer focus		Mechanism	Attitude	Primary Findings
	Influencer type	Context			
Park et al. (2021)	Source type (virtual vs. human); source number (singles vs. multiple)	Social media	Not discussed	Negative	Users tend to present negative feelings when interacting with virtual influencerscompared with human.
Arsenyan and Mirowska (2021)	Source type (human vs. human-like virtual vs. anime-like virtual)	Social media	Not discussed	Negative	Human-like virtual influencers receive significantly lower positive reactions from consumers than humans and anime-like virtual influencers.
Block and Lovegrove (2021)	Source type (virtual influencer)	Communication	Not discussed	Not discussed	The virtual influencer's identity and messaging effectively integrate into a single strategic communication tool.
Sands et al. (2022)	Source type (AI vs. human); source agency (autonomous vs. managed)	Social media	Social psychological distance; need for uniqueness	Both	Consumers perceive AI influencers and humans as equivalent in terms of following intention and perceived personalization, while AI influencers are perceived lowers source trust than human.
Thomas and Fowler (2021)	Source type (AI vs. celebrity)	Brand endorsement of transgressions	Not discussed	Positive	AI influencers can produce positive brand benefits like those produced by human celebrity endorsers.
Sung et al. (2022)	Source type (AI vs.none)	Mixed reality	Not discussed	Positive	Storytelling via an AI influencers can promote more effectively story immersion and story learning than without virtual influencers.
Stein et al. (2022)	Source type (virtual vs. human)	Video streaming	Human-likeness; perceived similarity	Both	Virtual influencers result in stronger parasocial interactions; however, this result is suppressed by the lower perception of human-likeness.
This article	Source type (virtual vs. human)	Endorsement effectiveness	Perceived sensory capability; perceived credibility	Negative	Virtual influencers decrease endorsement effect compared to human influencers through consumer's lower sensory perceptions and lower perceived credibility.

Appendix B

Theoretical Framework.

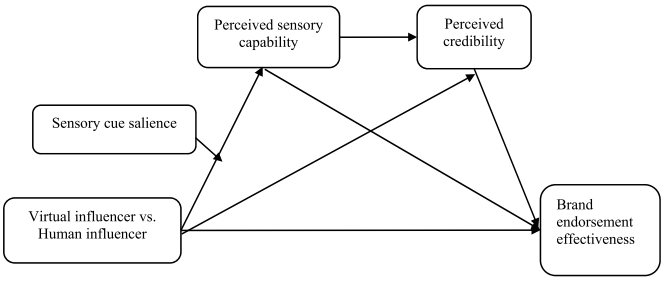


Fig. 1. Theoretical Framework

Appendix C

Study Measures.

Perceived sensory capability (Stafford et al., 2014; 1 = Strongly Disagree, 7 = Strongly Agree)
I perceived capacity of the Namey/Virtual_Namey to see.
I perceived capacity of the Namey/Virtual_Namey to hear.
I perceived capacity of the Namey/Virtual_Namey to touch.
I perceived capacity of the Namey/Virtual_Namey to smell.
I perceived capacity of the Namey/Virtual_Namey to taste.
Perceived credibility (Janssen et al., 2022; 1 = Strongly Disagree, 7 = Strongly Agree)

(*continued on next page*)

(*continued*)

I think Namey/Virtual_Namey is	Untrustworthy/trustworthy Insincere/sincere Unreliable/reliable
Brand attitude (Kim and Kim, 2021a; 1 = Strongly Disagree, 7 = Strongly Agree)	
How did you feel about the brand that was endorsed by Namey/Virtual_Namey?	Desirable/not desirable Pleasant/not pleasant Likable/not likable Good/bad
Purchase intention (Janssen et al., 2022; 1 = Strongly Disagree, 7 = Strongly Agree)	
The next time you are looking to purchase this type of product, how likely are you to buy this product?	

Appendix D

Experiment Figures.

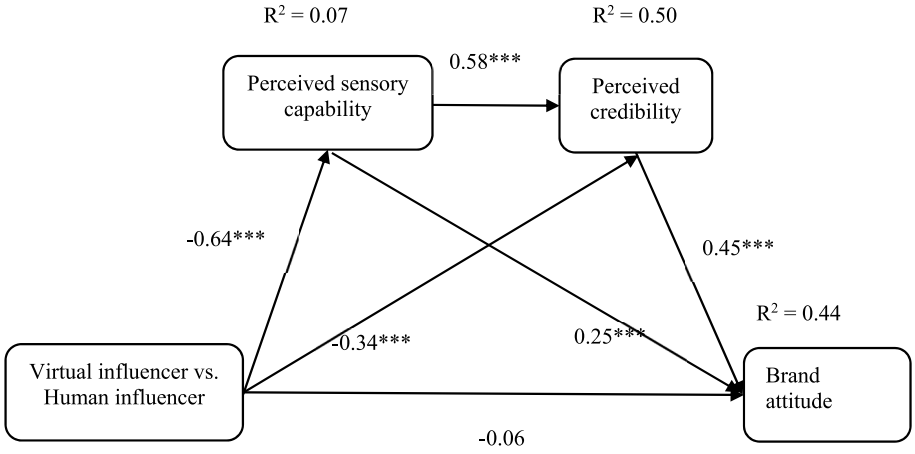


Fig. 2. Study 2a Results on Brand Attitude.

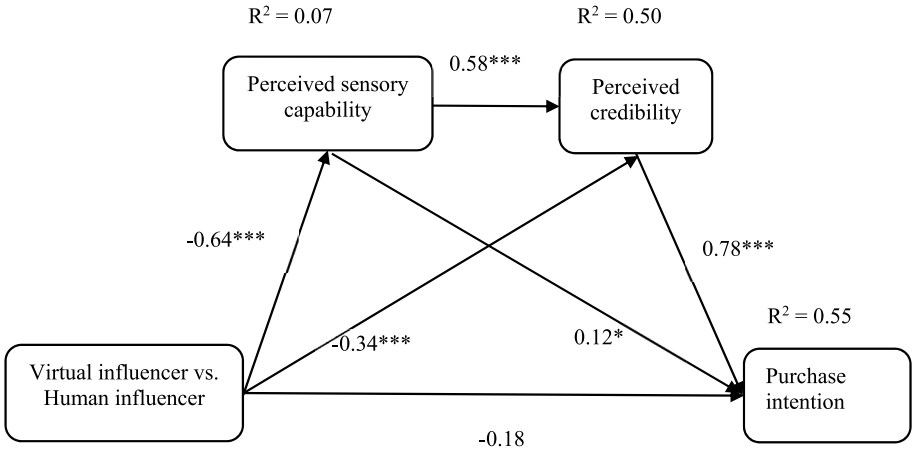


Fig. 3. Study 2a Results on Purchase Intention.

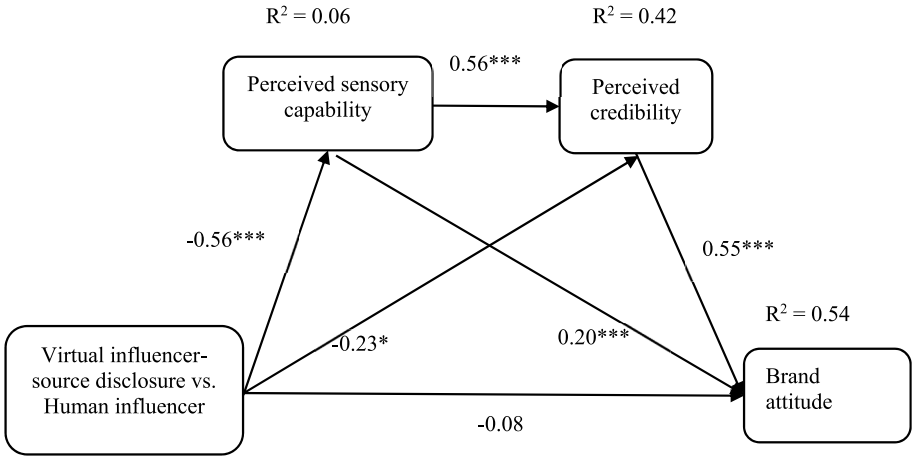


Fig. 4. Study 2b Results on Brand Attitude.

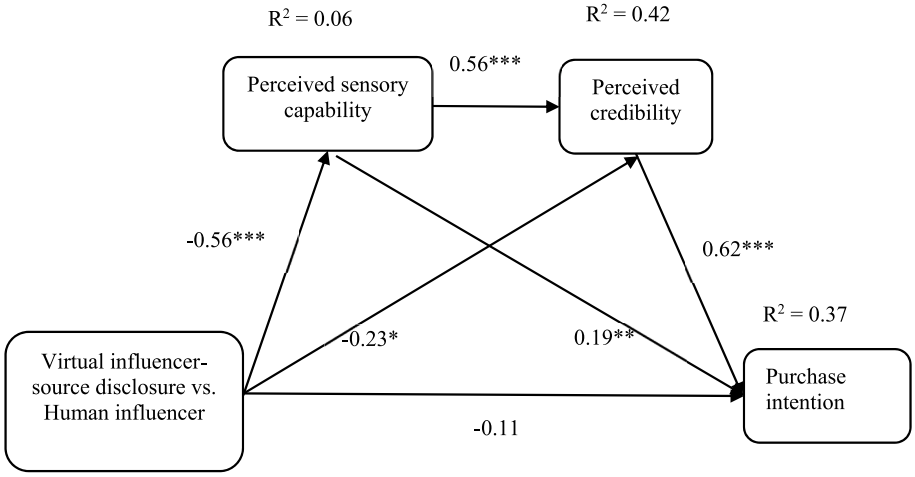


Fig. 5. Study 2b Results on Purchase Intention.

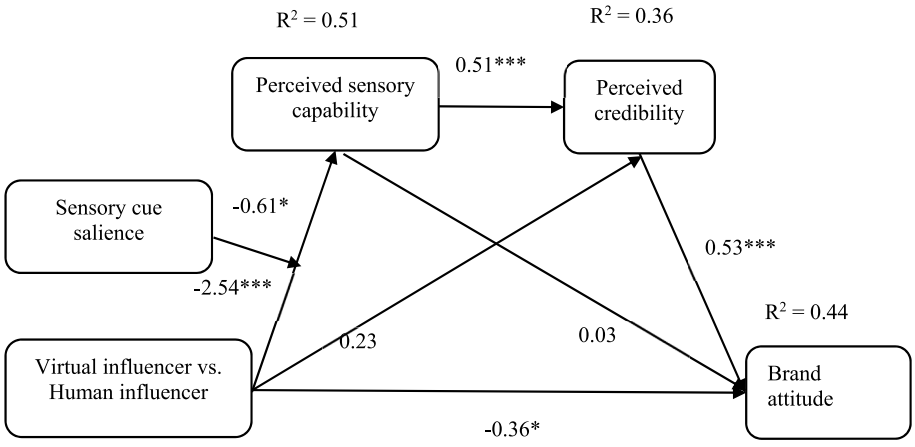


Fig. 6. Study 3 Results on Brand Attitude

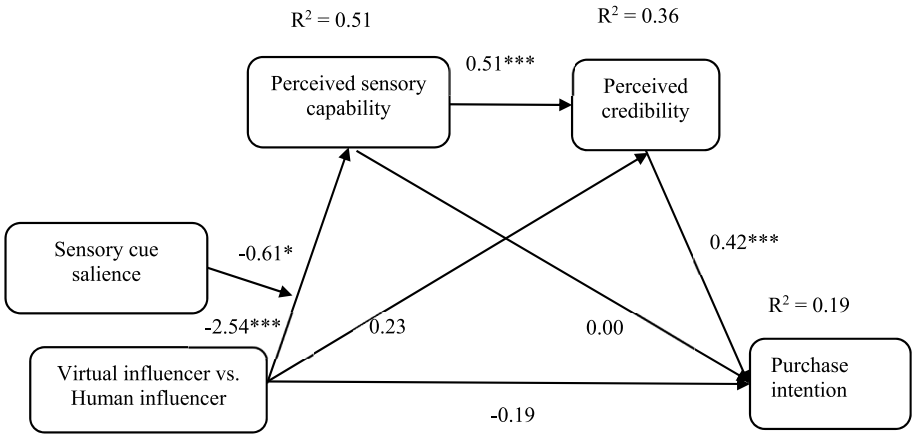
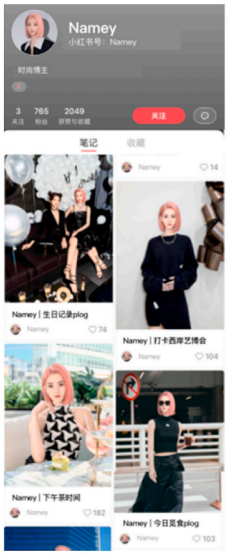


Fig. 7. Study 3 Results on Purchase Intention

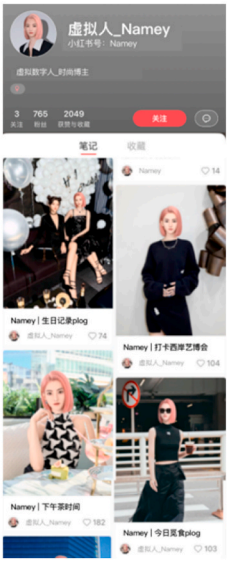
Appendix E

Experiment Stimuli in Study 2a and 2b.

I. Human influencer



II. Virtual influencer



Experiment Stimuli in Study 3.

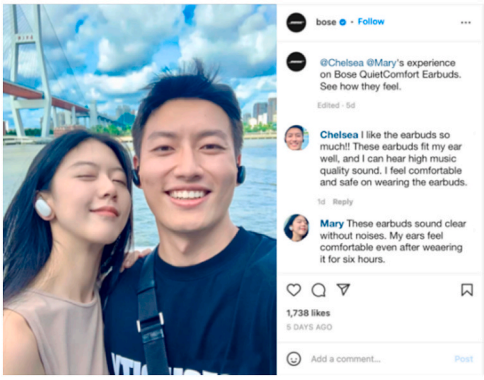
I. VI * High sensory cue salience



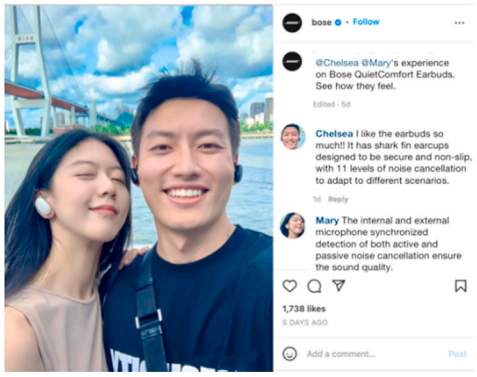
II. VI * Low sensory cue salience



III. HI * High sensory cue salience



IV. HI * Low sensory cue salience



Appendix F

Abbreviation Table.

Abbreviation	Full name
AI	Artificial Intelligence
CAGR	Compound Annual Growth Rate
LIWC	Linguistic Inquiry and Word Count
VI	Virtual influencer
HI	Human influencer
VI_source disclosure	Virtual influencer-source disclosure
VI_source concealed	Human influencer-Source concealed

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